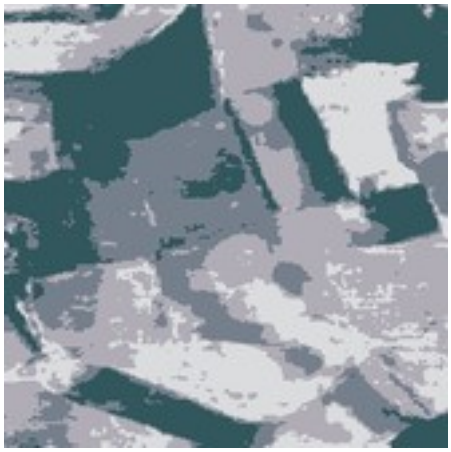
SAR Image Analysis Report

Original SAR (not found; preview shown)

Colorized



K-Means Segmented

Metadata:

- Original File: Not available
- Colorized File: colorized_1764305779.png
- Processed On: 2025-11-28 10:26:38 IST
- SAR Classifier Confidence: 0.000
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Water (Blue Tone) | Mean RGB: (219, 221, 223) | Area: 17.95%

Cluster 1: Likely Water (Blue Tone) | Mean RGB: (119, 128, 142) | Area: 24.81%

Cluster 2: Likely Water (Blue Tone) | Mean RGB: (177, 174, 183) | Area: 31.66%

Cluster 3: Mixed / Other Surface | Mean RGB: (50, 87, 92) | Area: 25.59%

Land Cover Summary:

- Likely Water (Blue Tone) → 17.95%
- Likely Water (Blue Tone) → 24.81%
- Likely Water (Blue Tone) → 31.66%
- Mixed / Other Surface → 25.59%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely water (blue tone) (74.42% combined).

Secondary presence: mixed / other surface (25.59%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is water (31.42%). Significant water presence suggests focusing on hydrological monitoring and flood risk assessment for the area. The second-largest class is mixed/other (25.4%), so combined management strategies addressing both water and mixed/other are advisable. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.000
Dominant Class Percentage: 31.66%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.